

Empiricist

14+3 · Course smoke · not ImageNet-scale

Architecture card + T6-T12 ablations + E1/E2/E6 smoke honesty. Do NOT claim 79.3% or 16×A100 from course runs.

Spine: Predict embeddings, not pixels · multi-block + EMA + narrow predictor · JEA|MAE|I-JEPA ·
→V-JEPA→2-AC (cite 2-AC for robots) · 54.2 / 66.9≫40.7 / ~10× / 72.4

Architecture card — paper defaults

Context / EMA target / predictor / loss / mask

Context encoder

ViT (B/L/H/G); visible patches only; no [CLS]; eval = avg-pool target-encoder

Target encoder

Same ViT; full image; EMA 0.996→1.0; mask OUTPUT not input (T11)

Predictor

Narrow ViT emb=384; depth 6/12/16 by backbone; positional mask tokens; M times

Multi-block mask

M=4 targets scale (0.15,0.2) aspect (0.75,1.5); context (0.85,1.0)×complement

Loss

Mean patch-wise L2 in embedding space; grads to θ, ϕ ; $\bar{\theta}$ EMA-only

Opt

AdamW BS2048; LR 1e-4→1e-3 (15ep)→1e-6 cos; WD 0.04→0.4; 224² default

Fig. 3 — architecture schematic

Context encoder · EMA target · narrow predictor · L2 emb

Generative Architectures. Reconstruction-based methods for self-supervised learning can also be cast in the framework of EBMs using Generative Architectures; see Figure 2b. Generative Architectures learn to directly reconstruct a signal y from a compatible signal x , using a decoder network that is conditioned on an additional (possibly latent) variable z to facilitate reconstruction. In the context of image-based pretraining, one common approach in computer vision is to produce compatible x, y pairs using masking [9, 38] where x is a copy of the image y , but with some of the patches masked. The conditioning variable z then corresponds to a set of (possibly learnable) mask and position tokens, that specifies to the decoder which image patches to reconstruct. Representation collapse is not a concern with these architectures as long as the informational capacity of z is low compared to the signal y .

Joint-Embedding Predictive Architectures. As shown in Figure 2c, Joint-Embedding Predictive Architectures [48]

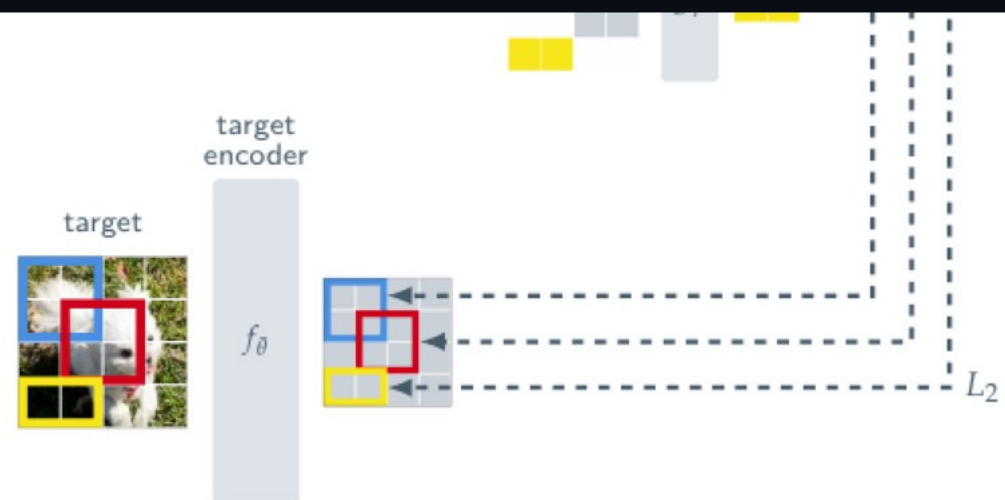


Figure 3. **I-JEPA.** The Image-based Joint-Embedding Predictive Architecture uses a single context block to predict the representations of various target blocks originating from the same image. The context encoder is a Vision Transformer (ViT), which only processes the visible context patches. The predictor is a narrow ViT that takes the context encoder output and, conditioned on positional tokens (shown in color), predicts the representations of a target block at a specific location. The target representations correspond to the outputs of the target-encoder, the weights of which are updated at each iteration via an exponential moving average of the context encoder weights.

Smoke reiml target: mask non-overlap, predictor shape = target patches, finite L2, EMA update.

T6 — multi-block is the thesis

ViT-B/16 · 300 ep · 1N-1% linear

multi-block

4× Block(0.15,0.2) · ctx (0.85,1.0)×comp

54.2

rasterized

3 quadrants · complement

15.5

block

1× Block(0.6) · complement

20.2

random

Random(0.6) · complement

17.6

T7 — emb >> pixels

Central JEPA vs generative claim

66.9

Target-encoder output
ViT-L/16 · 500 ep · 1% lin

40.7

Pixels
ViT-L/16 · 800 ep · 1% lin

Caveat: epochs differ (500 vs 800) — not perfectly controlled. Direction is the claim; smoke matches architecture+steps.

T8-T10 — scales & #targets

App. C · ViT-B/16 300ep · 1% lin

T8 target scale

(0.15,0.2)→54.2
too small 19.2
too large 33.6

T9 context scale

(0.40)→31.2
(0.65)→47.1
(0.85)→54.2

T10 #targets

1→9.0
2→22.0
3→48.5
4→54.2

T11 + T12 — output mask & predictor depth

Architecture knobs

T11 — where to mask

Output of target-encoder: 67.3

Input of target-encoder: 56.1

ViT-H/16 · 300 ep · 1% lin

Output masking → more semantic targets

T12 — predictor depth

depth 12 → 66.9

depth 6 → 64.0

ViT-L/16 · 500 ep · 1% lin Defaults: B→6, L/H→12, G→16; width 384 bottleneck helps (T14)

Also T14 width: 384 → 70.7 vs 1024 → 68.4 (1% FT). Narrow predictor is a feature.

Course smoke — E1 + E2 + E6 honesty

CIFAR / IN-100 · ViT-S — NOT paper absolute %

E1 P0 Mask: multi-block vs random vs block
Expect multi-block wins directionally — not 54.2

Y smoke

E2 P0 JEPA emb+EMA vs pixel decoder
Expect emb > pixels on semantic probe — not 66.9 vs 40.7

Y smoke

E6 P0 Frozen linear / k-NN on CIFAR100 or IN-100
Protocol first; absolute % $\ll$ paper H/14

Y smoke

E3 Target/context/#targets mini-grid
4–6 short runs after E1 scaffold

Y if GPU free

E10 ViT-H/14 IN-1K 300ep 16×A100
Cite paper only — never claim from course

N SKIP

GPU honesty — cite only vs smoke

Paper numbers are not course targets

CITE ONLY (do not re-run / do not claim from smoke)

ViT-H/14 · 16×A100 · <72 h wall · <1200 GPU-h · ~10× vs MAE H/14 · ~2.5× faster than iBOT ViT-S/16

COURSE SMOKE BUDGET

E1/E2 ViT-S/16 CIFAR100 BS256-512 · 1×24GB · 6-18 h · ≤40-60 GPU-h total for E1-E4+E6. Log train L2, probe top-1, mem, steps/sec.

Pass bar: paper-aligned TRENDS with GPU honesty logged — never 'we got 79.3%'.

Hypotheses — seat 4 pack for all seats

Falsifiable on tiny data

H-mask Multi-block still beats random/single-block on tiny linear/k-NN after matched short pretrain.

H-space Matched steps: EMA emb prediction beats pixel recon on semantic probes; pixel may win low-level toy.

H-predictor Too-shallow predictors underfit positional conditioning → worse probe; depth helps until wall.

H-transfer Smoke probe beats scratch supervised tiny baseline; absolute % will NOT match paper H/14.

H-ckpt E8 Public I-JEPA ViT features competitive with MAE on household robot stills — optional BEHAVIOR bridge.

Success bar vs paper claims

What Empiricist must / must not deliver

Paper claim	Local success	NOT expected
T6 multi-block critical	multi > random & block on tiny probe	Matching 54.2%
T7 emb » pixels	emb+EMA > pixel on semantic probe	Matching 66.9 vs 40.7
T12 deeper predictor	depth ↑ improves or plateaus	Need depth-12 on ViT-L
No-aug strong linear	smoke > scratch; discuss DINO via E8	79.3% H/14
16×A100 <72h	Cite only	Re-running it

Fig. 4 — multi-block defaults to freeze

Freeze these before sweeping other axes



Figure 4. Examples of our context and target-masking strategy. Given an image, we randomly sample 4 target blocks with scale in the range (0.15, 0.2) and aspect ratio in the range (0.75, 1.5). Next, we randomly sample a context block with scale in the range (0.85, 1.0) and remove any overlapping target blocks. Under this

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Empiricist rule: lock mask defaults from paper, then ablate one axis per Slurm array task.

Numbers tattoo — pin these

Shared across all five seats

79.3%

ViT-H/14 linear

73.3%

H/14 · 1% labels

81.1% / 77.3%

H/16@448 lin / 1%

54.2

multi-block (T6)

66.9 >> 40.7

emb >> pixels (T7)

72.4

Clevr Dist (T4)

~10×

cheaper than MAE H/14

<72 h

16×A100 H/14

<1200

GPU-hours H/14

87.1 vs 87.8

FT H/16@448 vs MAE

no view augs

single-view JEPA

2-AC not here

robots → cite 2506.09985

Empiricist — close

Takeaways + exam bite

1. Architecture: context / EMA target / narrow predictor / L2 emb / multi-block defaults.

2. Ablations: T6 54.2 · T7 66.9>>40.7 · T8-10 · T11 · T12 — cite paper absolute %.

3. Smoke: E1+E2+E6 on CIFAR/IN-100 ViT-S — directional only.

4. GPU: <1200 GPU-h / 16×A100 <72h = cite only — never claim from course.

EXAM BITE

Smoke validates mask + emb-vs-pixel DIRECTION; paper owns 79.3% and 16×A100.